

EEGLAB Practical Exercise 01

Session: EEG Data Import • Channel Location • Bad Channel Detection • Filtering

Name & Family Name

Date

Hamid Reza Ghahramani

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PART 1 | Import & Data Organization

1. Import the data (64 channels, sampling rate=2048) and save the corresponding EEGLAB dataset with channel locations described in Standard-10-20-Cap64.ced, and save it as project01_loc.set.

PART 2 | Bad Channel Detection & Correction

2. Remove any bad channels if they exist and replace them with interpolated ones, and re-reference the dataset to average reference, resampled to sampling rate=128, and save the new dataset as project01_loc_badrmd_avg_128.set.

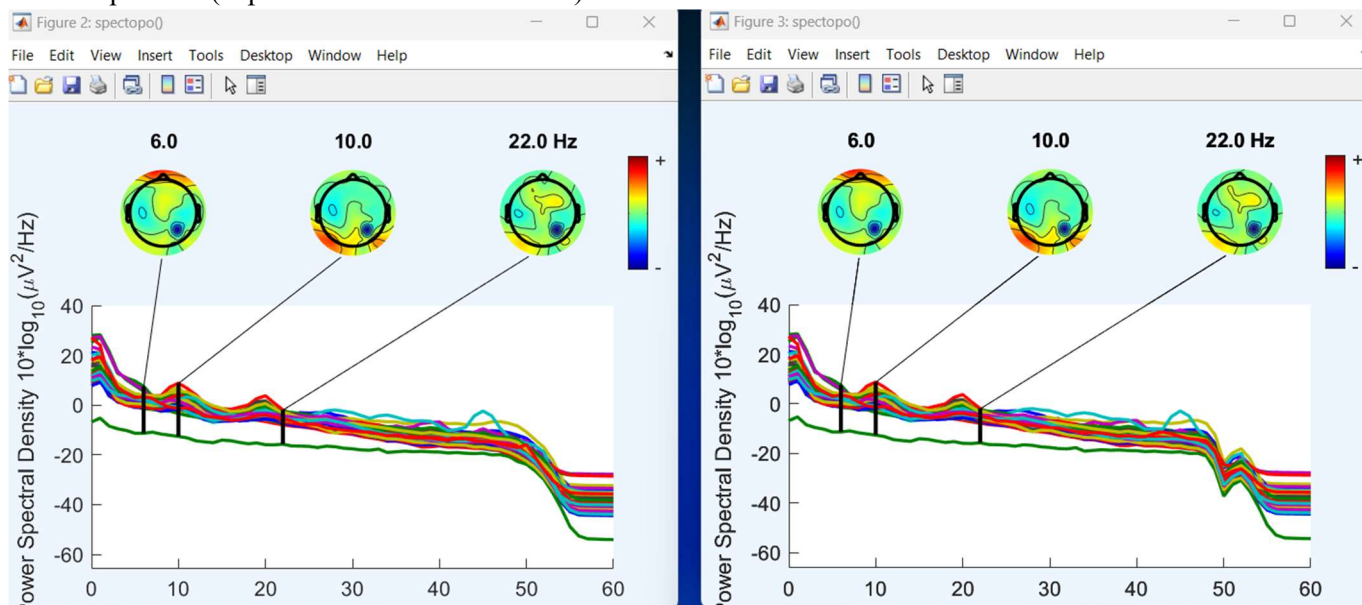
PART 3 | Filtering Pipeline

- 3.1 Filter the line noise by the zapline plugin and then use a high-pass filter of 0.5 Hz and a low-pass filter of 45 Hz, and save the new dataset as project01_loc_badrmd_avg_128_filtered.set.

Notch filter versus Zapline

- 3.2 Create a second copy of the dataset processed with Zapline. Compare it with the 50-Hz notch-filter version. Explain which method reduced line noise more clearly and why a 45-Hz low-pass filter affects the interpretation of

this comparison. (explain it and take screen shot)



The original dataset shows a clear peak around 50 Hz, which is caused by power-line noise. After applying the 50-Hz notch filter, the peak is removed, but a clear dip or notch appears around 50 Hz. In contrast, the Zapline-processed dataset shows a much smoother and more uniform spectrum around 50 Hz, without a clear peak or dip. Therefore, in our data, Zapline produced a smoother removal of the line noise than the 50-Hz notch filter.

The 45-Hz low-pass filter affects the comparison because it strongly reduces frequencies above 45 Hz, including the 50-Hz line noise. Therefore, after applying the 45-Hz low-pass filter, the difference between Zapline and the 50-Hz notch filter cannot be clearly evaluated. The comparison is more meaningful before applying the low-pass filter.

PART 4 | Artifact rejection

4. Perform an artifact rejection for the last dataset by scrolling the data, and save the new dataset as .set

You need to send the last file as [YourName]_project01_loc_badrmd_avg_128_filtered_manrejected.set